EXPERIMENT -5

AIM- Create SQL views for department salary summary and employee hierarchy. Test updatability of views. Implement a recursive CTE to display reporting chains.

OBJECTIVES

1. To understand the concept and purpose of SQL views.
2. To create a view for displaying department-wise salary summaries.
3. To create a view representing the employee-manager hierarchy.
4. To test whether the created views are updatable.
5. To understand the limitations of updating views containing aggregate functions and joins.
6. To understand Common Table Expressions (CTEs).
7. To implement a recursive CTE for traversing employee-manager relationships.
8. To display complete reporting chains from employees to the top-level manager.

THEORY

SQL View

A **view** is a virtual table created from the result of a SQL query. It does not normally store the actual data separately; instead, it stores the query definition used to retrieve the data from underlying tables.

Views are useful for:

- Simplifying complex queries.
- Providing a customized representation of data.
- Restricting access to specific columns or rows.
- Improving data abstraction.
- Reusing frequently required queries.

Department Salary Summary View

A department salary summary view combines employee and department information and uses aggregate functions such as COUNT() and AVG() to provide summarized salary information for each department.

For example, it can display:

- Department name
- Number of employees
- Total salary
- Average salary
- Maximum salary
- Minimum salary

Because this view contains **aggregate functions and GROUP BY**, it is generally not directly updatable.

Employee Hierarchy View

An employee hierarchy represents the relationship between an employee and their manager.

For this purpose, the Employee table contains a ManagerID attribute that refers back to EmpID in the same table.

Updatable View

An updatable view is a view through which operations such as INSERT, UPDATE, or DELETE can be performed on the underlying table.

A simple view based on a single table is generally more likely to be updatable.

Non-Updatable Views

A view may not be directly updatable when its definition contains features that make it difficult to map the result back to individual base-table rows.

Common examples include views containing:

- Aggregate functions
- GROUP BY
- DISTINCT
- Set operations
- Certain joins
- Calculated or derived results

Therefore, a department salary summary containing SUM(), AVG(), and GROUP BY cannot normally be directly updated.

Similarly, the employee hierarchy view uses a self-join, so it should not be treated as a simple updatable view.

Common Table Expression

A **Common Table Expression (CTE)** is a temporary named result set that can be referenced within a SQL statement.

It is defined using the WITH clause.

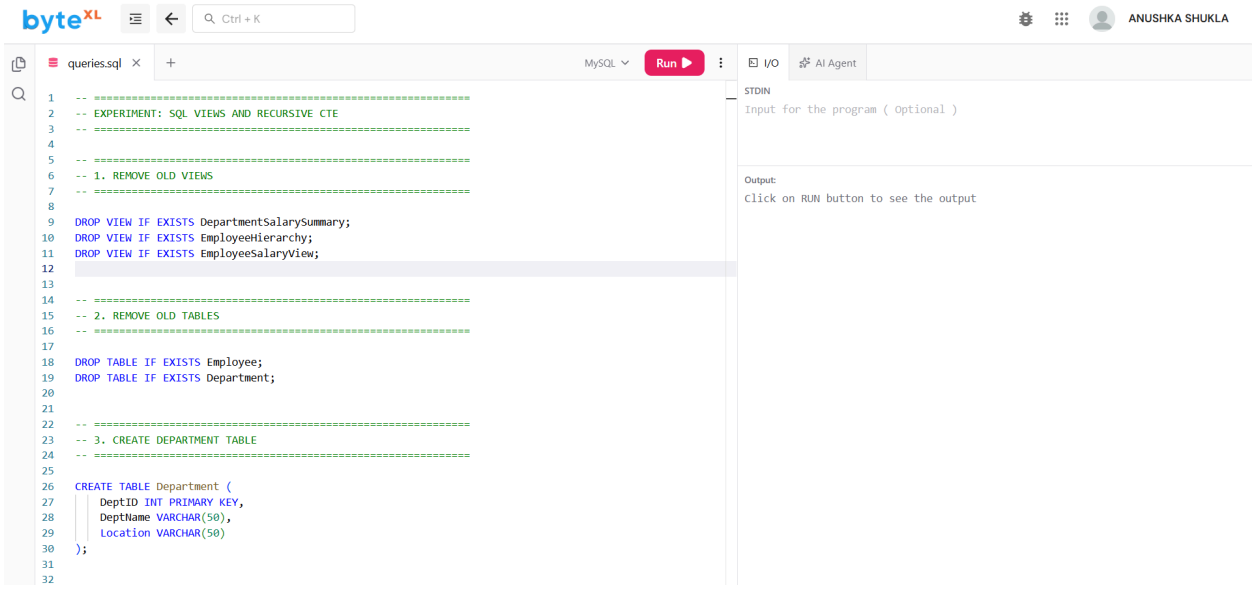
Recursive CTE

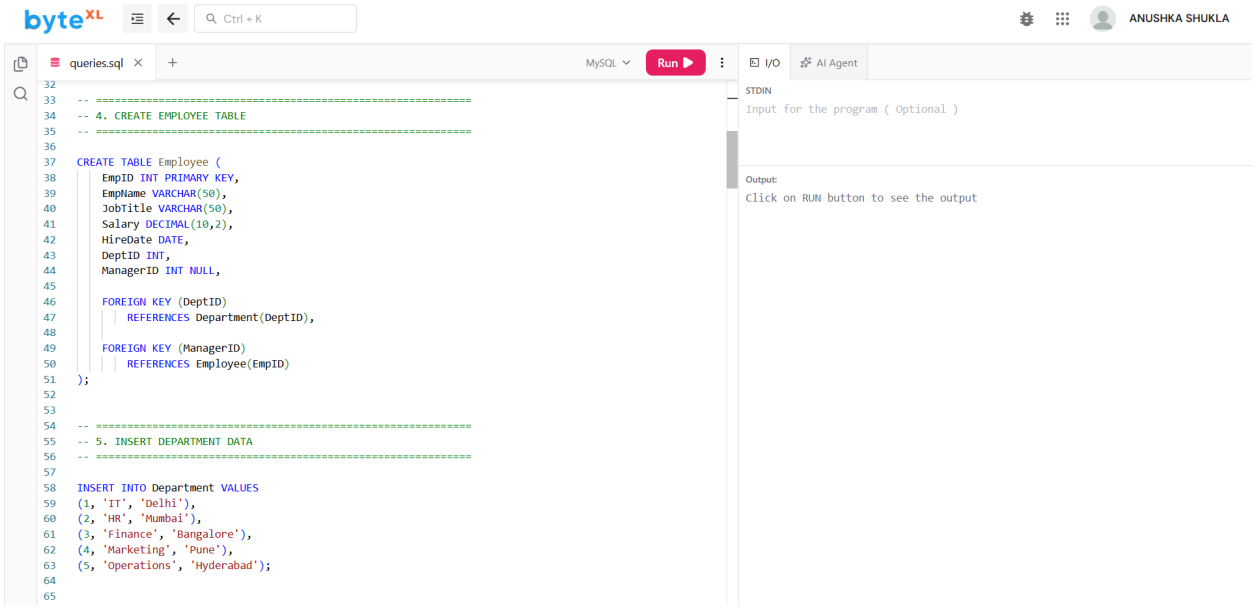
A **recursive CTE** is a CTE that refers to itself. It is useful for processing hierarchical or tree-structured data.

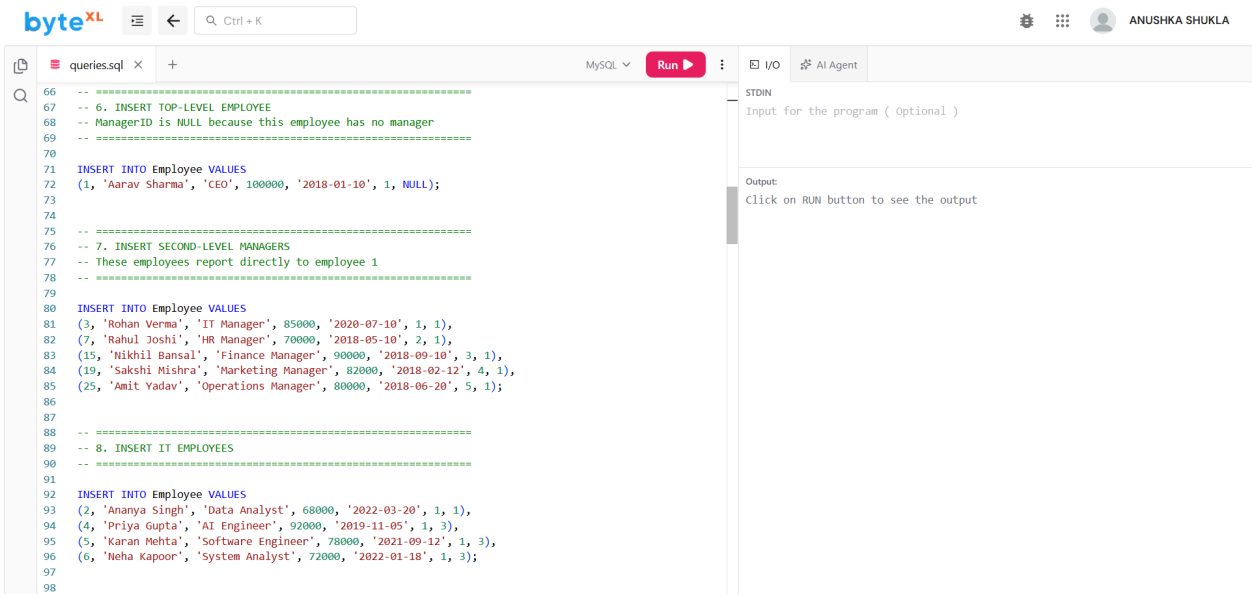
A recursive CTE normally consists of two parts:

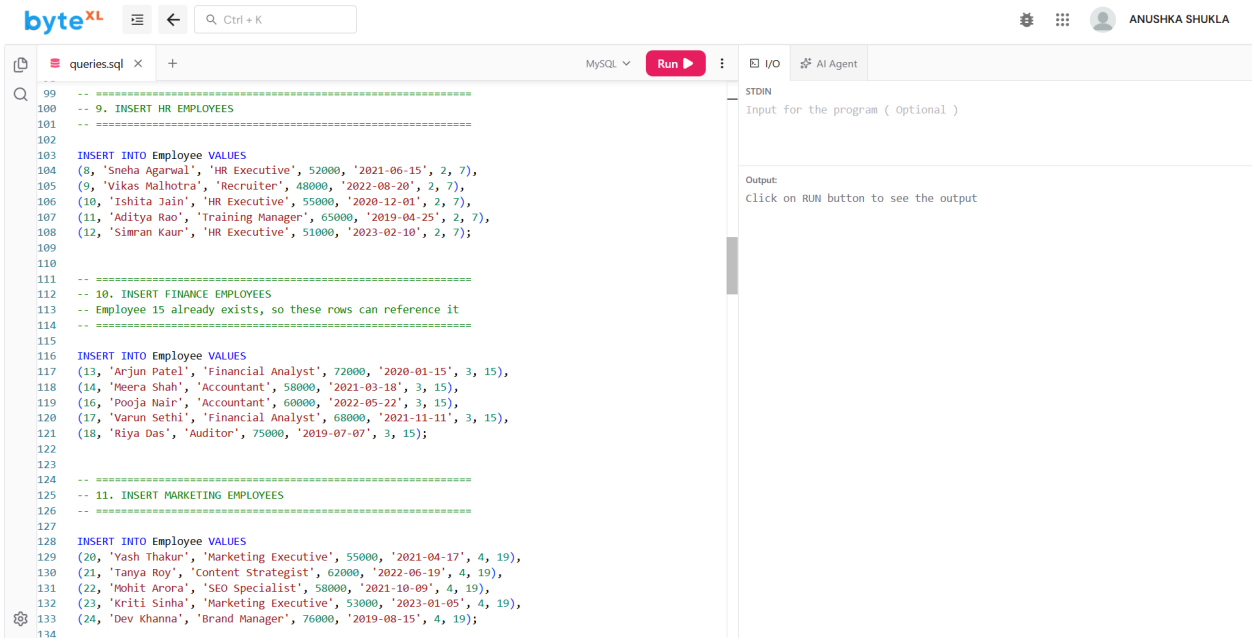
1. **Anchor member**: identifies the starting records.
2. **Recursive member**: repeatedly finds the next level of related records.

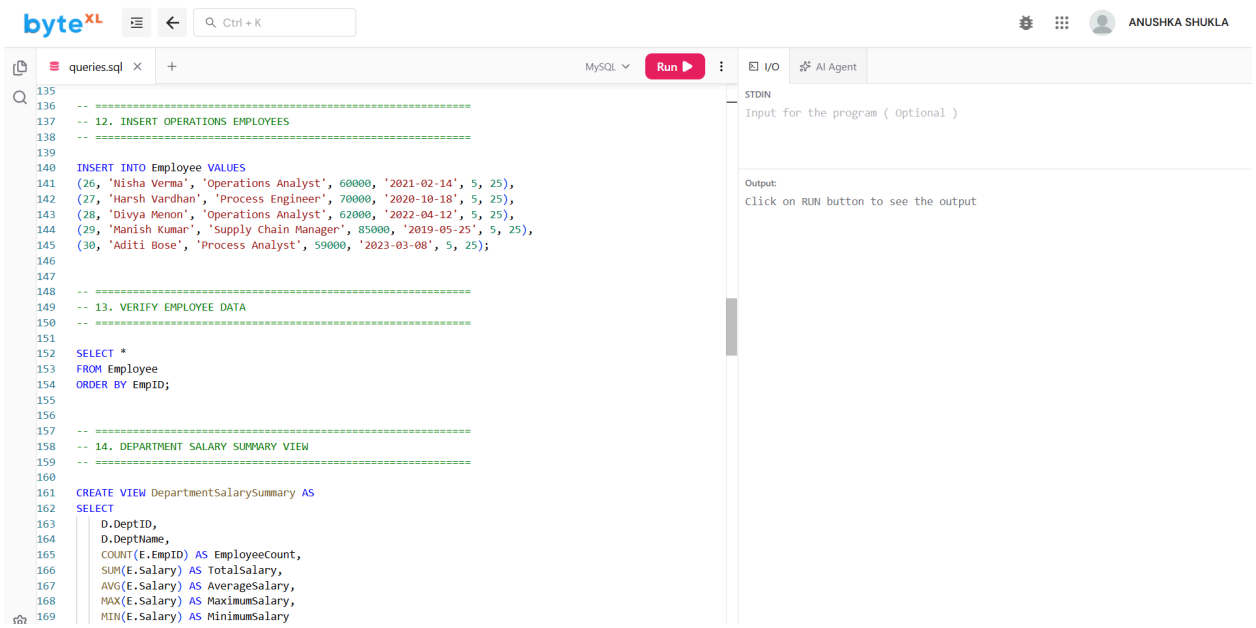
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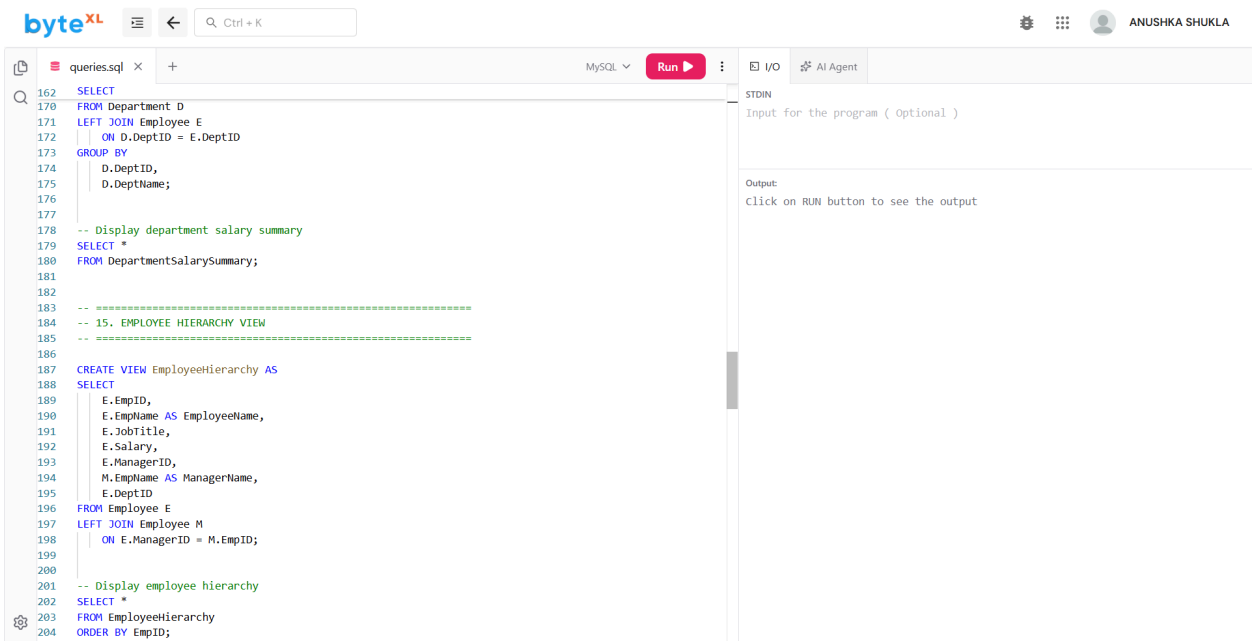

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-- ==
-- EXPERIMENT: SQL VIEWS AND RECURSIVE CTE
-- ==
-- ==
-- 1. REMOVE OLD VIEWS
-- ==
DROP VIEW IF EXISTS DepartmentSalarySummary;
DROP VIEW IF EXISTS EmployeeHierarchy;
DROP VIEW IF EXISTS EmployeeSalaryView;
-- ==
-- 2. REMOVE OLD TABLES
-- ==
DROP TABLE IF EXISTS Employee;
DROP TABLE IF EXISTS Department;
-- ==
-- 3. CREATE DEPARTMENT TABLE
-- ==
CREATE TABLE Department (
DeptID INT PRIMARY KEY,
DeptName VARCHAR(50),
Location VARCHAR(50)
);

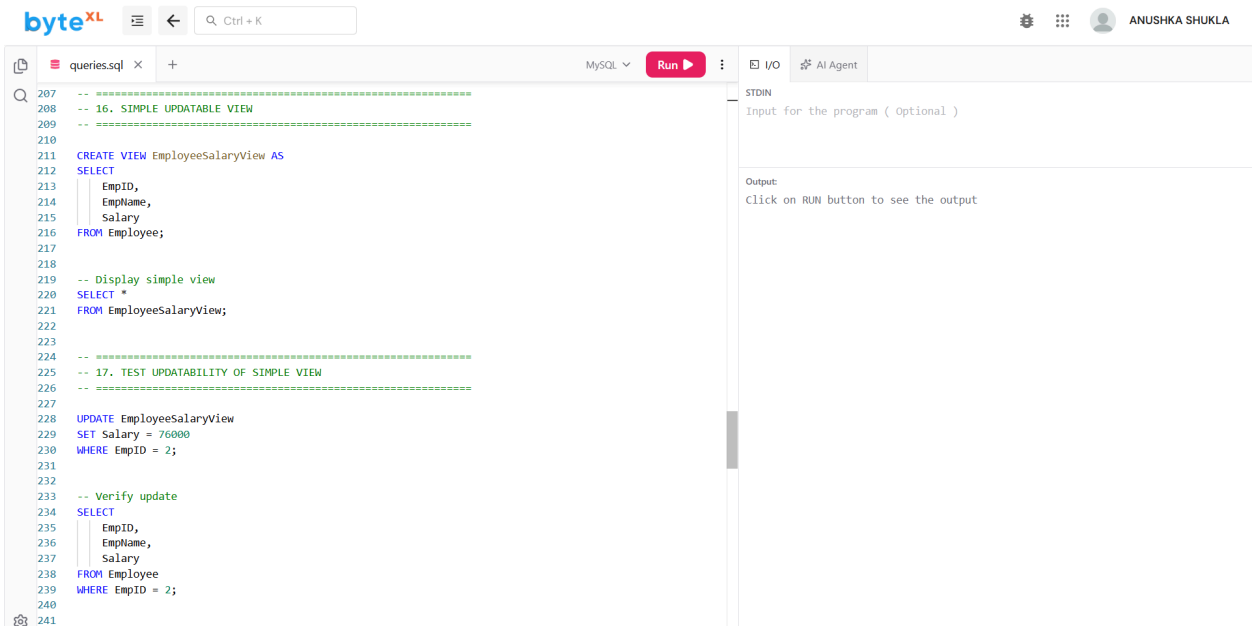

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-- 4. CREATE EMPLOYEE TABLE
-- ==
CREATE TABLE Employee (
EmpID INT PRIMARY KEY,
EmpName VARCHAR(50),
JobTitle VARCHAR(50),
Salary DECIMAL(10,2),
HireDate DATE,
DeptID INT,
ManagerID INT NULL,
FOREIGN KEY (DeptID)
REFERENCES Department(DeptID),
FOREIGN KEY (ManagerID)
REFERENCES Employee(EmpID)
);
-- ==
-- 5. INSERT DEPARTMENT DATA
-- ==
INSERT INTO Department VALUES
(1, 'IT', 'Delhi'),
(2, 'HR', 'Mumbai'),
(3, 'Finance', 'Bangalore'),
(4, 'Marketing', 'Pune'),
(5, 'Operations', 'Hyderabad');

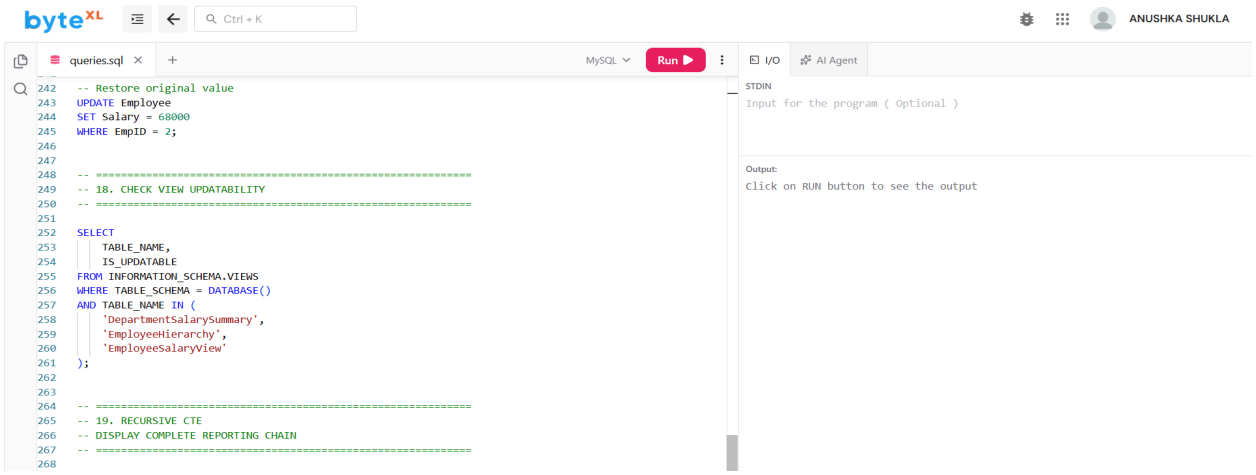

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Output:
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-- ==
-- 6. INSERT TOP-LEVEL EMPLOYEE
-- ManagerID is NULL because this employee has no manager
-- ==
INSERT INTO Employee VALUES
(1, 'Aarav Sharma', 'CEO', 100000, '2018-01-10', 1, NULL);
-- ==
-- 7. INSERT SECOND-LEVEL MANAGERS
-- These employees report directly to employee 1
-- ==
INSERT INTO Employee VALUES
(3, 'Rohan Verma', 'IT Manager', 85000, '2020-07-10', 1, 1),
(7, 'Rahul Joshi', 'HR Manager', 70000, '2018-05-10', 2, 1),
(15, 'Nikhil Bansal', 'Finance Manager', 90000, '2018-09-10', 3, 1),
(19, 'Sakshi Mishra', 'Marketing Manager', 82000, '2018-02-12', 4, 1),
(25, 'Amit Yadav', 'Operations Manager', 80000, '2018-06-20', 5, 1);
-- ==
-- 8. INSERT IT EMPLOYEES
-- ==
INSERT INTO Employee VALUES
(2, 'Ananya Singh', 'Data Analyst', 68000, '2022-03-20', 1, 1),
(4, 'Priya Gupta', 'AI Engineer', 92000, '2019-11-05', 1, 3),
(5, 'Karan Mehta', 'Software Engineer', 78000, '2021-09-12', 1, 3),
(6, 'Neha Kapoor', 'System Analyst', 72000, '2022-01-18', 1, 3);

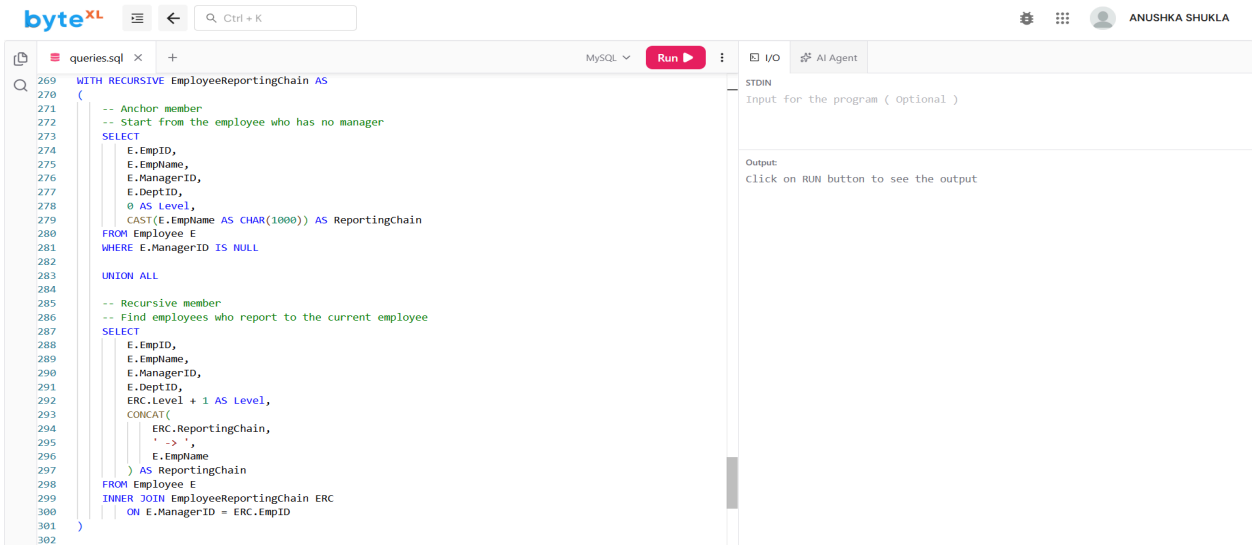

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Output:
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-- ==
-- 9. INSERT HR EMPLOYEES
-- ==
INSERT INTO Employee VALUES
(8, 'Sneha Agarwal', 'HR Executive', 52000, '2021-06-15', 2, 7),
(9, 'Vikas Malhotra', 'Recruiter', 48000, '2022-08-20', 2, 7),
(10, 'Ishita Jain', 'HR Executive', 55000, '2020-12-01', 2, 7),
(11, 'Aditya Rao', 'Training Manager', 65000, '2019-04-25', 2, 7),
(12, 'Simran Kaur', 'HR Executive', 51000, '2023-02-10', 2, 7);
-- ==
-- 10. INSERT FINANCE EMPLOYEES
-- Employee 15 already exists, so these rows can reference it
-- ==
INSERT INTO Employee VALUES
(13, 'Arjun Patel', 'Financial Analyst', 72000, '2020-01-15', 3, 15),
(14, 'Meera Shah', 'Accountant', 58000, '2021-03-18', 3, 15),
(16, 'Pooja Nair', 'Accountant', 60000, '2022-05-22', 3, 15),
(17, 'Varun Sethi', 'Financial Analyst', 68000, '2021-11-11', 3, 15),
(18, 'Riya Das', 'Auditor', 75000, '2019-07-07', 3, 15);
-- ==
-- 11. INSERT MARKETING EMPLOYEES
-- ==
INSERT INTO Employee VALUES
(20, 'Yash Thakur', 'Marketing Executive', 55000, '2021-04-17', 4, 19),
(21, 'Tanya Roy', 'Content Strategist', 62000, '2022-06-19', 4, 19),
(22, 'Mohit Arora', 'SEO Specialist', 58000, '2021-10-09', 4, 19),
(23, 'Kriti Sinha', 'Marketing Executive', 53000, '2023-01-05', 4, 19),
(24, 'Dev Khanna', 'Brand Manager', 76000, '2019-08-15', 4, 19);

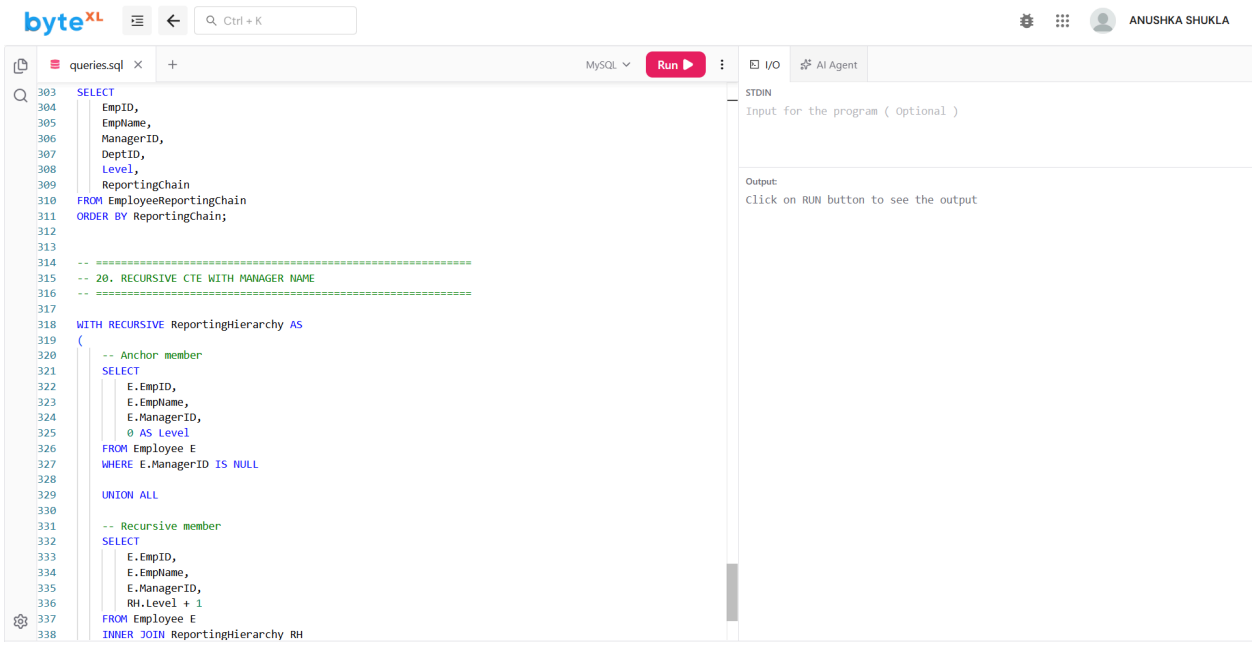

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Output:
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-- ==
-- 12. INSERT OPERATIONS EMPLOYEES
-- ==
INSERT INTO Employee VALUES
(26, 'Nisha Verma', 'Operations Analyst', 60000, '2021-02-14', 5, 25),
(27, 'Harsh Vardhan', 'Process Engineer', 70000, '2020-10-18', 5, 25),
(28, 'Divya Menon', 'Operations Analyst', 62000, '2022-04-12', 5, 25),
(29, 'Manish Kumar', 'Supply Chain Manager', 85000, '2019-05-25', 5, 25),
(30, 'Aditi Bose', 'Process Analyst', 59000, '2023-03-08', 5, 25);
-- ==
-- 13. VERIFY EMPLOYEE DATA
-- ==
SELECT *
FROM Employee
ORDER BY EmpID;
-- ==
-- 14. DEPARTMENT SALARY SUMMARY VIEW
-- ==
CREATE VIEW DepartmentSalarySummary AS
SELECT
D.DeptID,
D.DeptName,
COUNT(E.EmpID) AS EmployeeCount,
SUM(E.Salary) AS TotalSalary,
AVG(E.Salary) AS AverageSalary,
MAX(E.Salary) AS MaximumSalary,
MIN(E.Salary) AS MinimumSalary

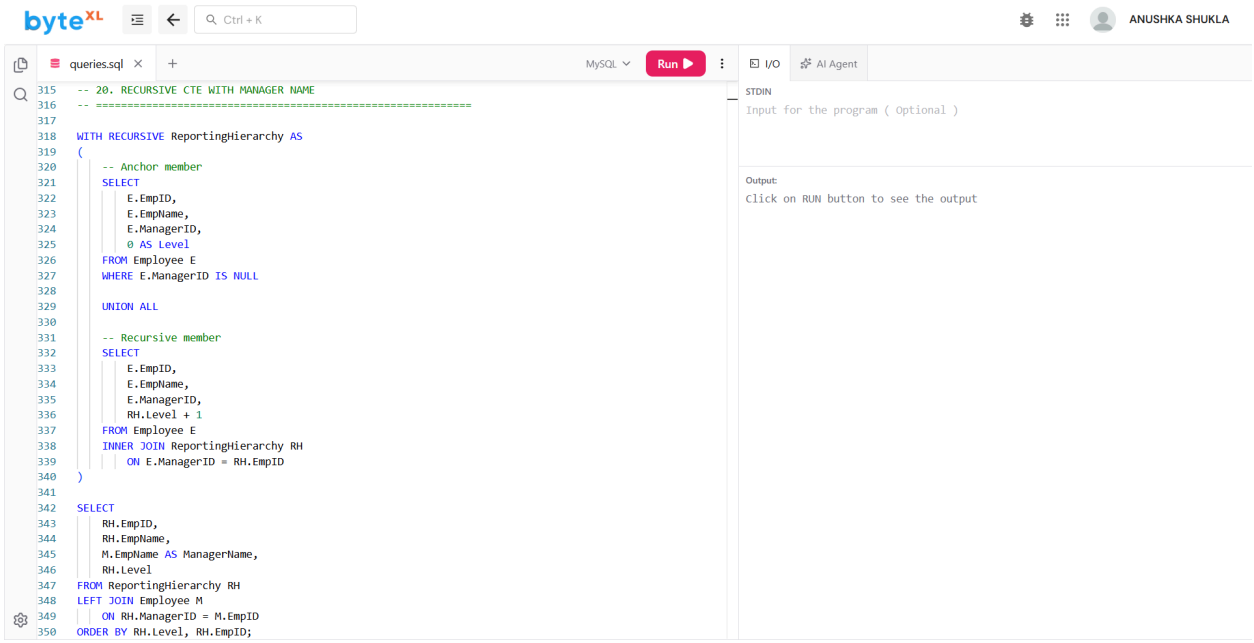

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Output:
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SELECT
FROM Department D
LEFT JOIN Employee E
ON D.DeptID = E.DeptID
GROUP BY
D.DeptID,
D.DeptName;
-- Display department salary summary
SELECT *
FROM DepartmentSalarySummary;
-- ==
-- 15. EMPLOYEE HIERARCHY VIEW
-- ==
CREATE VIEW EmployeeHierarchy AS
SELECT
E.EmpID,
E.EmpName AS EmployeeName,
E.JobTitle,
E.Salary,
E.ManagerID,
M.EmpName AS ManagerName,
E.DeptID
FROM Employee E
LEFT JOIN Employee M
ON E.ManagerID = M.EmpID;
-- Display employee hierarchy
SELECT *
FROM EmployeeHierarchy
ORDER BY EmpID;


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Input for the program (Optional)
Output:
Click on RUN button to see the output
-- ==
-- 16. SIMPLE UPDATABLE VIEW
-- ==
CREATE VIEW EmployeeSalaryView AS
SELECT
EmpID,
EmpName,
Salary
FROM Employee;
-- Display simple view
SELECT *
FROM EmployeeSalaryView;
-- ==
-- 17. TEST UPDATABILITY OF SIMPLE VIEW
-- ==
UPDATE EmployeeSalaryView
SET Salary = 76000
WHERE EmpID = 2;
-- Verify update
SELECT
EmpID,
EmpName,
Salary
FROM Employee
WHERE EmpID = 2;

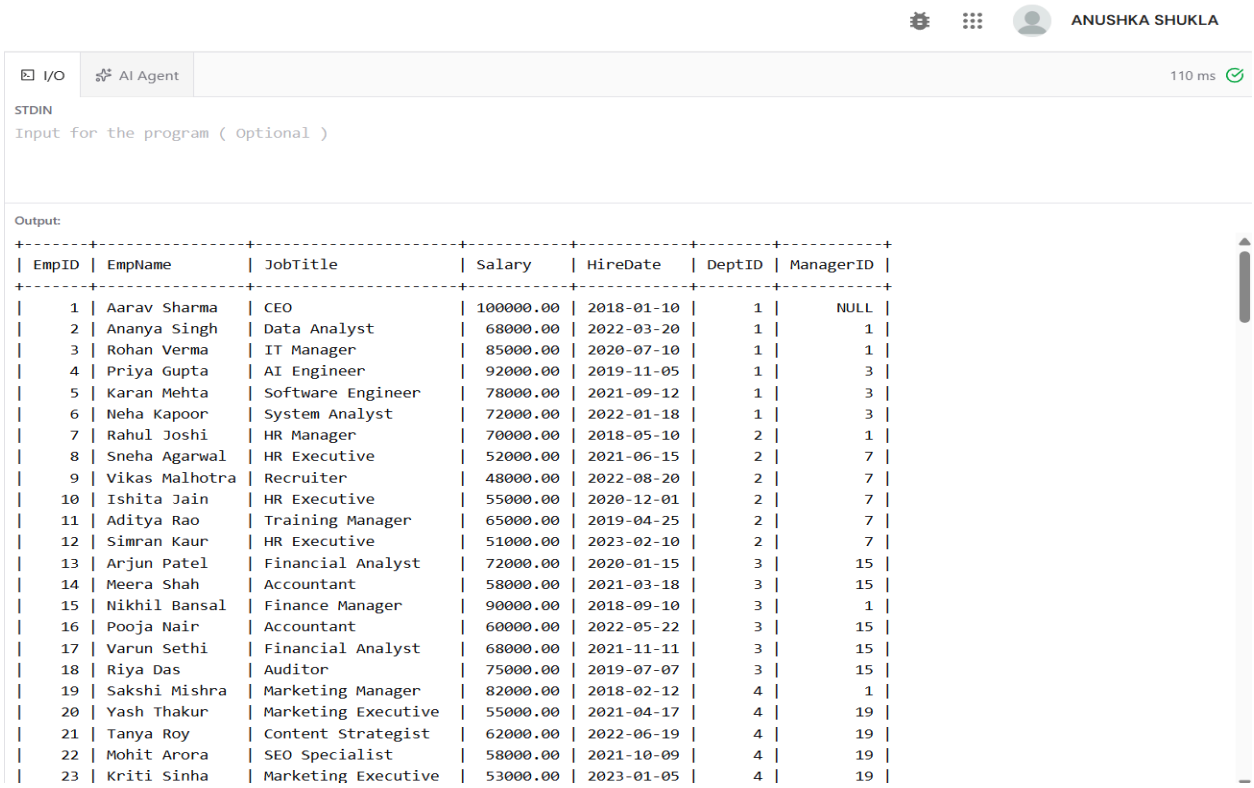

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Input for the program (Optional)
Output:
Click on RUN button to see the output
-- Restore original value
UPDATE Employee
SET Salary = 68000
WHERE EmpID = 2;
-- ==
-- 18. CHECK VIEW UPDATABILITY
-- ==
SELECT
TABLE_NAME,
IS_UPDATABLE
FROM INFORMATION_SCHEMA.VIEWS
WHERE TABLE_SCHEMA = DATABASE()
AND TABLE_NAME IN (
'DepartmentSalarySummary',
'EmployeeHierarchy',
'EmployeeSalaryView'
);
-- ==
-- 19. RECURSIVE CTE
-- DISPLAY COMPLETE REPORTING CHAIN
-- ==


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Output:
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WITH RECURSIVE EmployeeReportingChain AS
(
-- Anchor member
-- Start from the employee who has no manager
SELECT
E.EmpID,
E.EmpName,
E.ManagerID,
E.DeptID,
0 AS Level,
CAST(E.EmpName AS CHAR(1000)) AS ReportingChain
FROM Employee E
WHERE E.ManagerID IS NULL
UNION ALL
-- Recursive member
-- Find employees who report to the current employee
SELECT
E.EmpID,
E.EmpName,
E.ManagerID,
E.DeptID,
ERC.Level + 1 AS Level,
CONCAT(
ERC.ReportingChain,
' -> ',
E.EmpName
) AS ReportingChain
FROM Employee E
INNER JOIN EmployeeReportingChain ERC
ON E.ManagerID = ERC.EmpID
)

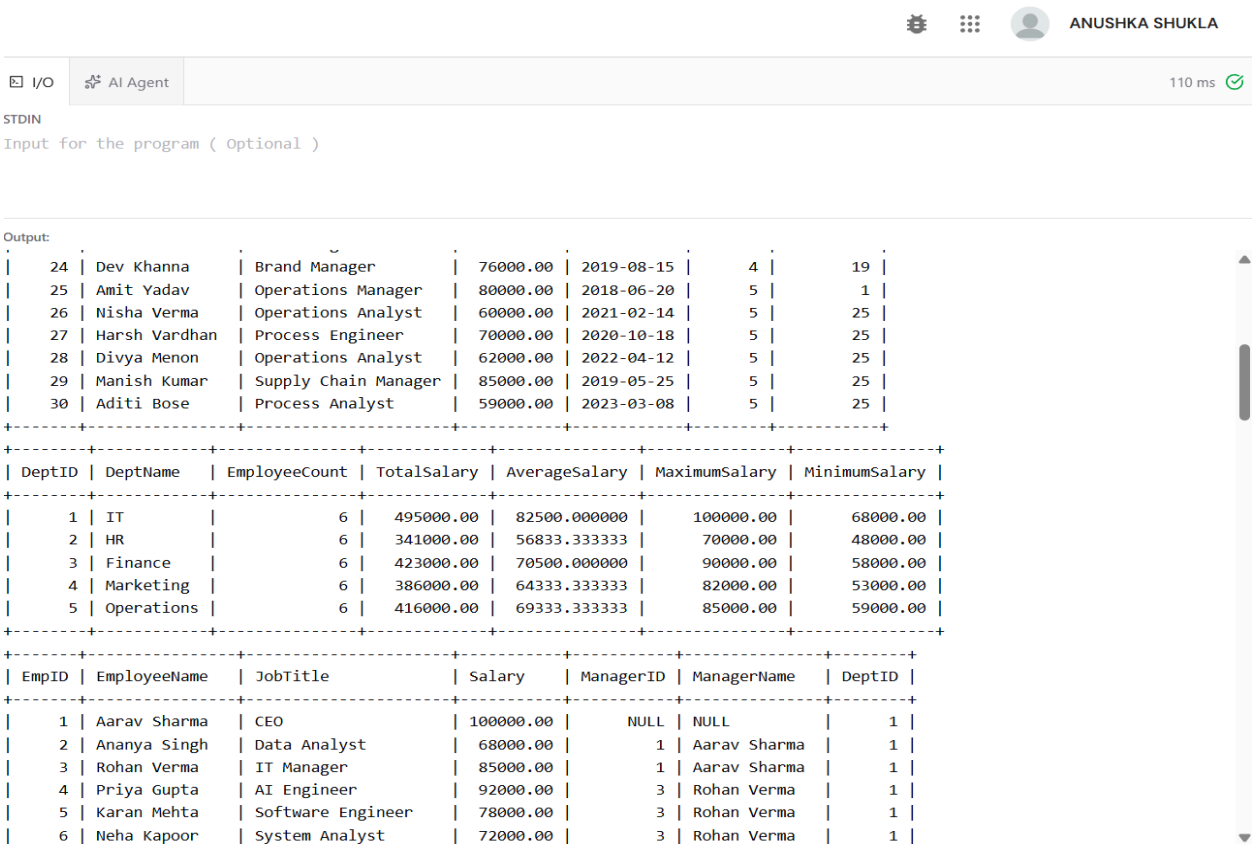

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Input for the program (Optional)
Output:
Click on RUN button to see the output
SELECT
EmpID,
EmpName,
ManagerID,
DeptID,
Level,
ReportingChain
FROM EmployeeReportingChain
ORDER BY ReportingChain;
-- ==
-- 20. RECURSIVE CTE WITH MANAGER NAME
-- ==
WITH RECURSIVE ReportingHierarchy AS
(
-- Anchor member
SELECT
E.EmpID,
E.EmpName,
E.ManagerID,
0 AS Level
FROM Employee E
WHERE E.ManagerID IS NULL
UNION ALL
-- Recursive member
SELECT
E.EmpID,
E.EmpName,
E.ManagerID,
RH.Level + 1
FROM Employee E
INNER JOIN ReportingHierarchy RH


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Input for the program (Optional)
Output:
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-- 20. RECURSIVE CTE WITH MANAGER NAME
-- ==
WITH RECURSIVE ReportingHierarchy AS
(
-- Anchor member
SELECT
E.EmpID,
E.EmpName,
E.ManagerID,
0 AS Level
FROM Employee E
WHERE E.ManagerID IS NULL
UNION ALL
-- Recursive member
SELECT
E.EmpID,
E.EmpName,
E.ManagerID,
RH.Level + 1
FROM Employee E
INNER JOIN ReportingHierarchy RH
ON E.ManagerID = RH.EmpID
)
SELECT
RH.EmpID,
RH.EmpName,
M.EmpName AS ManagerName,
RH.Level
FROM ReportingHierarchy RH
LEFT JOIN Employee M
ON RH.ManagerID = M.EmpID
ORDER BY RH.Level, RH.EmpID;

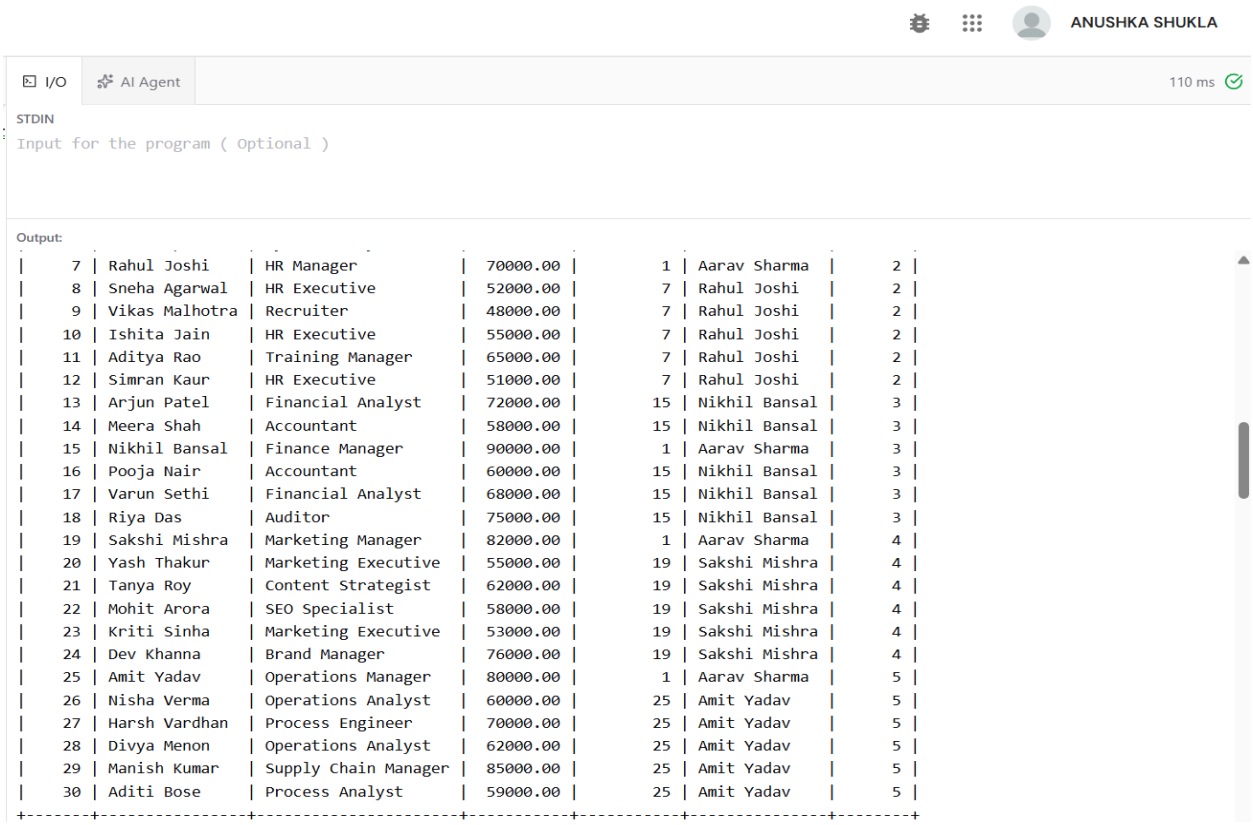
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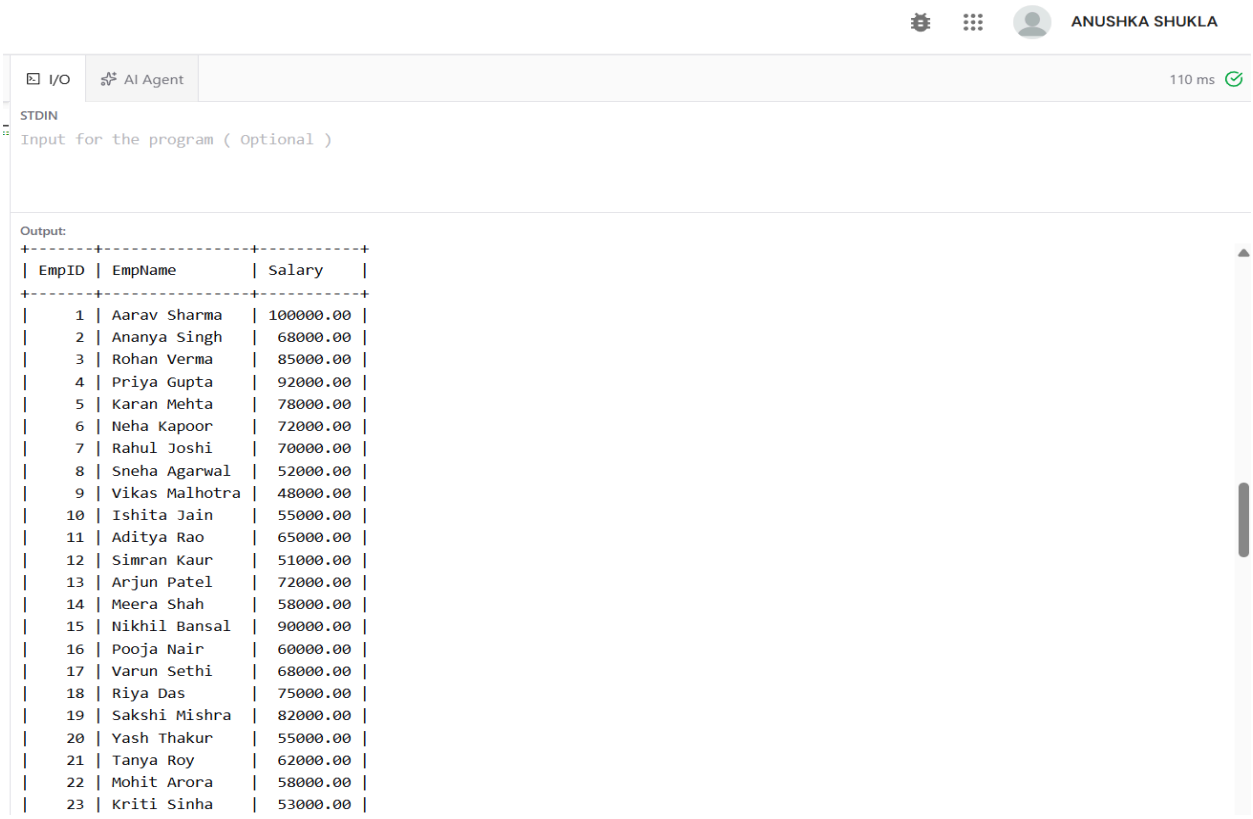


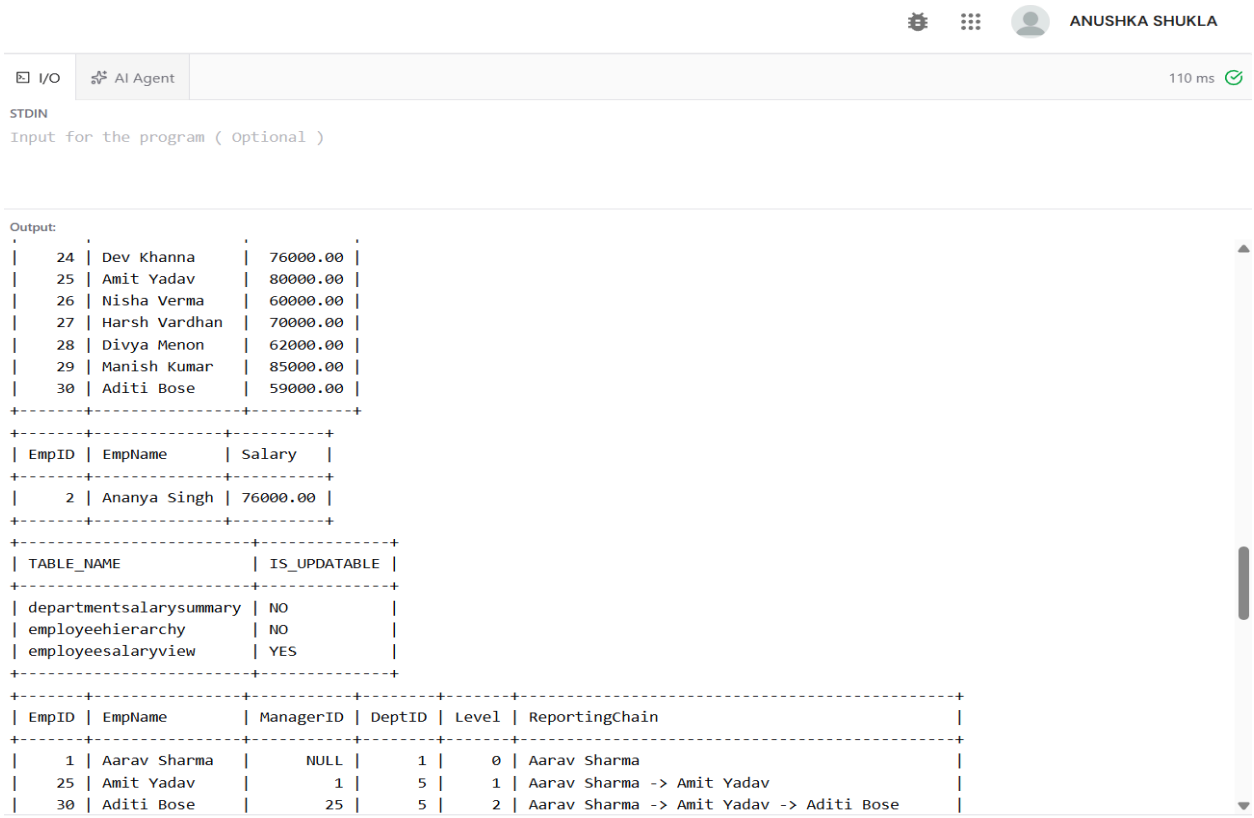

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+-------+----------------+----------------------+-----------+------------+--------+-----------+
| EmpID | EmpName        | JobTitle             | Salary    | HireDate   | DeptID | ManagerID |
+-------+----------------+----------------------+-----------+------------+--------+-----------+
|     1 | Aarav Sharma   | CEO                  | 100000.00 | 2018-01-10 |      1 |      NULL |
|     2 | Ananya Singh   | Data Analyst         |  68000.00 | 2022-03-20 |      1 |         1 |
|     3 | Rohan Verma    | IT Manager           |  85000.00 | 2020-07-10 |      1 |         1 |
|     4 | Priya Gupta    | AI Engineer          |  92000.00 | 2019-11-05 |      1 |         3 |
|     5 | Karan Mehta    | Software Engineer    |  78000.00 | 2021-09-12 |      1 |         3 |
|     6 | Neha Kapoor    | System Analyst       |  72000.00 | 2022-01-18 |      1 |         3 |
|     7 | Rahul Joshi    | HR Manager           |  70000.00 | 2018-05-10 |      2 |         1 |
|     8 | Sneha Agarwal  | HR Executive         |  52000.00 | 2021-06-15 |      2 |         7 |
|     9 | Vikas Malhotra | Recruiter            |  48000.00 | 2022-08-20 |      2 |         7 |
|    10 | Ishita Jain    | HR Executive         |  55000.00 | 2020-12-01 |      2 |         7 |
|    11 | Aditya Rao     | Training Manager     |  65000.00 | 2019-04-25 |      2 |         7 |
|    12 | Simran Kaur    | HR Executive         |  51000.00 | 2023-02-10 |      2 |         7 |
|    13 | Arjun Patel    | Financial Analyst    |  72000.00 | 2020-01-15 |      3 |        15 |
|    14 | Meera Shah     | Accountant           |  58000.00 | 2021-03-18 |      3 |        15 |
|    15 | Nikhil Bansal  | Finance Manager      |  90000.00 | 2018-09-10 |      3 |         1 |
|    16 | Pooja Nair     | Accountant           |  60000.00 | 2022-05-22 |      3 |        15 |
|    17 | Varun Sethi    | Financial Analyst    |  68000.00 | 2021-11-11 |      3 |        15 |
|    18 | Riya Das       | Auditor              |  75000.00 | 2019-07-07 |      3 |        15 |
|    19 | Sakshi Mishra  | Marketing Manager    |  82000.00 | 2018-02-12 |      4 |         1 |
|    20 | Yash Thakur    | Marketing Executive  |  55000.00 | 2021-04-17 |      4 |        19 |
|    21 | Tanya Roy      | Content Strategist   |  62000.00 | 2022-06-19 |      4 |        19 |
|    22 | Mohit Arora    | SEO Specialist       |  58000.00 | 2021-10-09 |      4 |        19 |
|    23 | Kriti Sinha    | Marketing Executive  |  53000.00 | 2023-01-05 |      4 |        19 |
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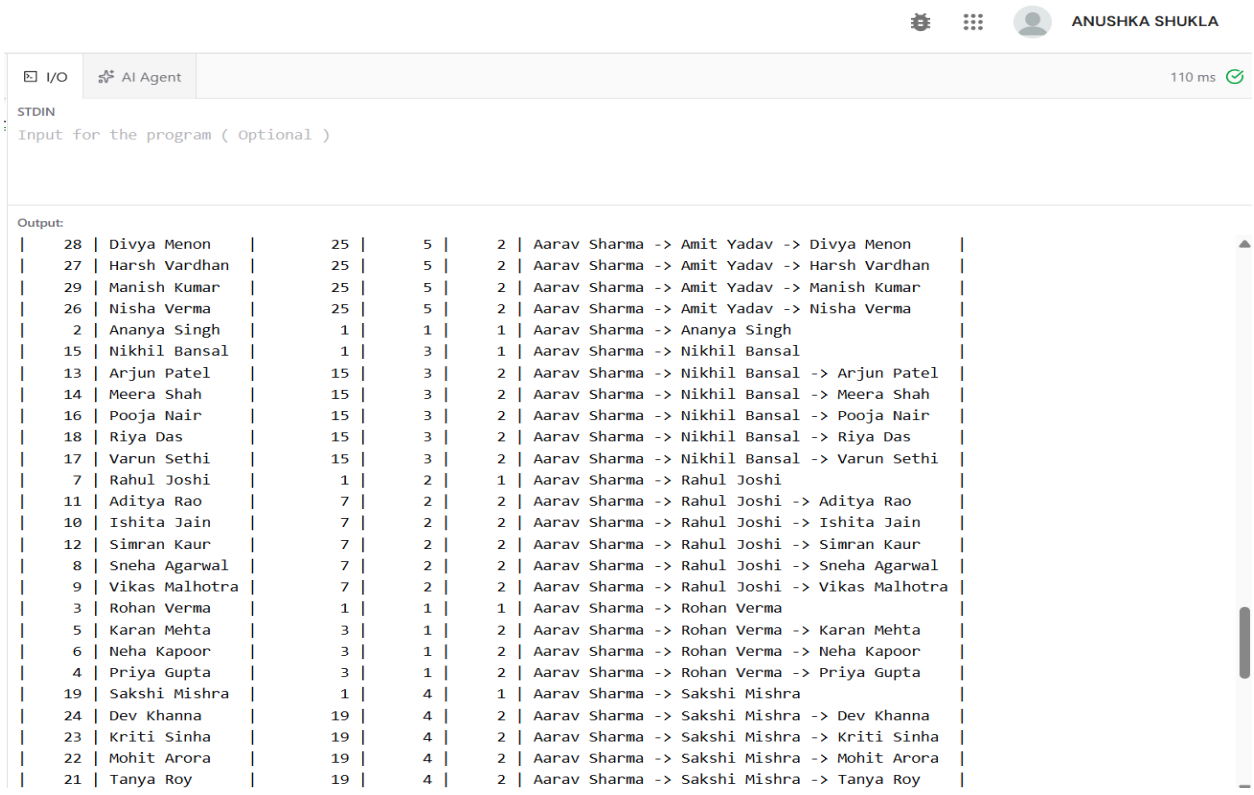


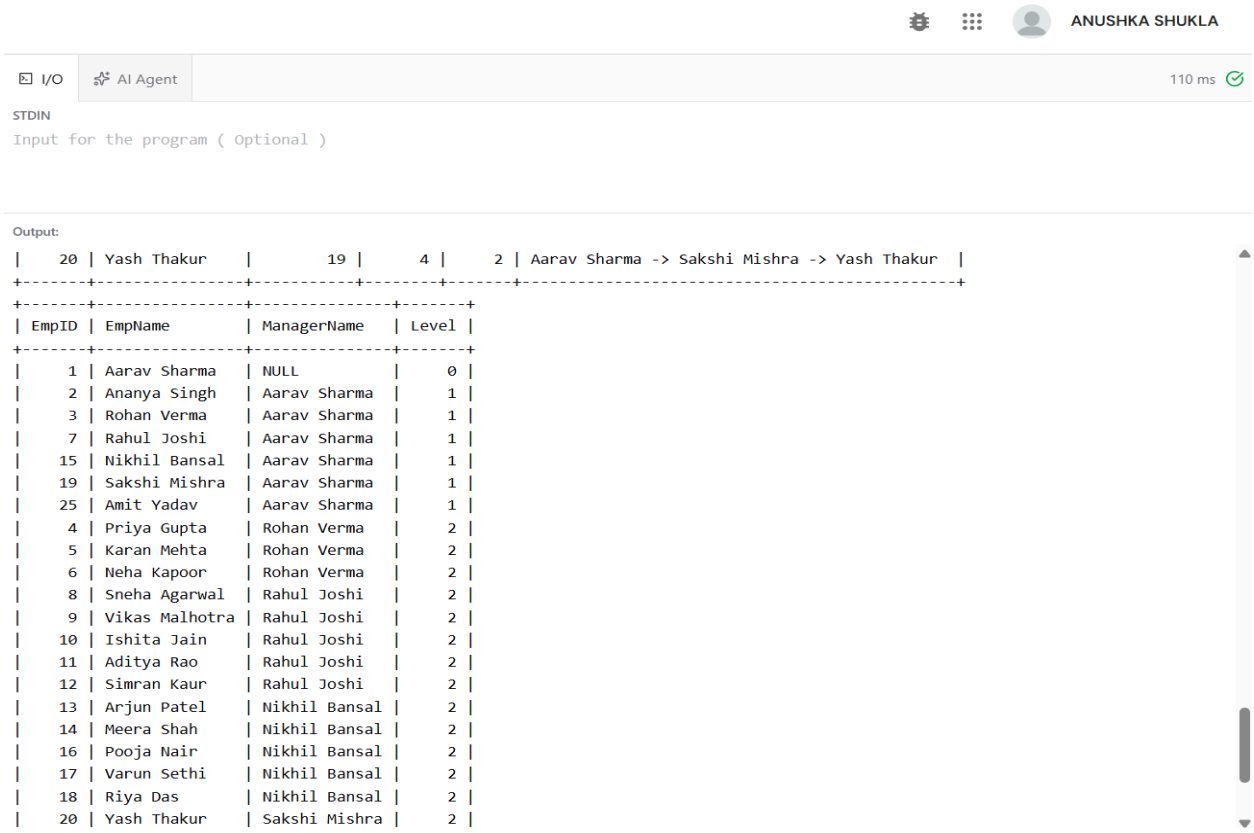

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|    24 | Dev Khanna     | Brand Manager        |  76000.00 | 2019-08-15 |      4 |        19 |
|    25 | Amit Yadav     | Operations Manager   |  80000.00 | 2018-06-20 |      5 |         1 |
|    26 | Nisha Verma    | Operations Analyst   |  60000.00 | 2021-02-14 |      5 |        25 |
|    27 | Harsh Vardhan  | Process Engineer     |  70000.00 | 2020-10-18 |      5 |        25 |
|    28 | Divya Menon    | Operations Analyst   |  62000.00 | 2022-04-12 |      5 |        25 |
|    29 | Manish Kumar   | Supply Chain Manager |  85000.00 | 2019-05-25 |      5 |        25 |
|    30 | Aditi Bose     | Process Analyst      |  59000.00 | 2023-03-08 |      5 |        25 |
+-------+----------------+----------------------+-----------+------------+--------+-----------+
+--------+------------+---------------+-------------+---------------+---------------+---------------+
| DeptID | DeptName   | EmployeeCount | TotalSalary | AverageSalary | MaximumSalary | MinimumSalary |
+--------+------------+---------------+-------------+---------------+---------------+---------------+
|      1 | IT         |             6 |   495000.00 |  82500.000000 |     100000.00 |      68000.00 |
|      2 | HR         |             6 |   341000.00 |  56833.333333 |      70000.00 |      48000.00 |
|      3 | Finance    |             6 |   423000.00 |  70500.000000 |      90000.00 |      58000.00 |
|      4 | Marketing  |             6 |   386000.00 |  64333.333333 |      82000.00 |      53000.00 |
|      5 | Operations |             6 |   416000.00 |  69333.333333 |      85000.00 |      59000.00 |
+--------+------------+---------------+-------------+---------------+---------------+---------------+
+-------+----------------+----------------------+-----------+-----------+---------------+--------+
| EmpID | EmployeeName   | JobTitle             | Salary    | ManagerID | ManagerName   | DeptID |
+-------+----------------+----------------------+-----------+-----------+---------------+--------+
|     1 | Aarav Sharma   | CEO                  | 100000.00 |      NULL | NULL          |      1 |
|     2 | Ananya Singh   | Data Analyst         |  68000.00 |         1 | Aarav Sharma  |      1 |
|     3 | Rohan Verma    | IT Manager           |  85000.00 |         1 | Aarav Sharma  |      1 |
|     4 | Priya Gupta    | AI Engineer          |  92000.00 |         3 | Rohan Verma   |      1 |
|     5 | Karan Mehta    | Software Engineer    |  78000.00 |         3 | Rohan Verma   |      1 |
|     6 | Neha Kapoor    | System Analyst       |  72000.00 |         3 | Rohan Verma   |      1 |
```

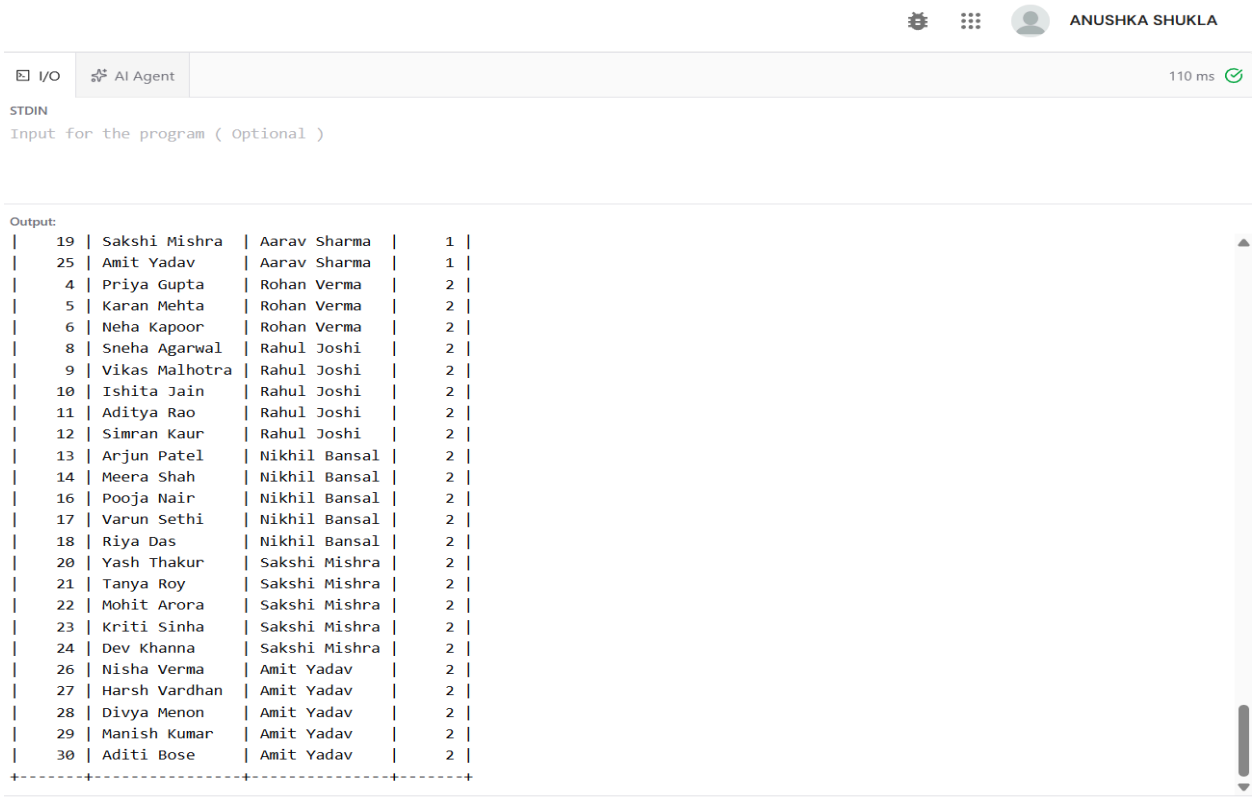

ANUSHKA SHUKLA
I/O
AI Agent
110 ms
STDIN
Input for the program (Optional)
Output:
7	Rahul Joshi	HR Manager	70000.00	1	Aarav Sharma	2
8	Sneha Agarwal	HR Executive	52000.00	7	Rahul Joshi	2
9	Vikas Malhotra	Recruiter	48000.00	7	Rahul Joshi	2
10	Ishita Jain	HR Executive	55000.00	7	Rahul Joshi	2
11	Aditya Rao	Training Manager	65000.00	7	Rahul Joshi	2
12	Simran Kaur	HR Executive	51000.00	7	Rahul Joshi	2
13	Arjun Patel	Financial Analyst	72000.00	15	Nikhil Bansal	3
14	Meera Shah	Accountant	58000.00	15	Nikhil Bansal	3
15	Nikhil Bansal	Finance Manager	90000.00	1	Aarav Sharma	3
16	Pooja Nair	Accountant	60000.00	15	Nikhil Bansal	3
17	Varun Sethi	Financial Analyst	68000.00	15	Nikhil Bansal	3
18	Riya Das	Auditor	75000.00	15	Nikhil Bansal	3
19	Sakshi Mishra	Marketing Manager	82000.00	1	Aarav Sharma	4
20	Yash Thakur	Marketing Executive	55000.00	19	Sakshi Mishra	4
21	Tanya Roy	Content Strategist	62000.00	19	Sakshi Mishra	4
22	Mohit Arora	SEO Specialist	58000.00	19	Sakshi Mishra	4
23	Kriti Sinha	Marketing Executive	53000.00	19	Sakshi Mishra	4
24	Dev Khanna	Brand Manager	76000.00	19	Sakshi Mishra	4
25	Amit Yadav	Operations Manager	80000.00	1	Aarav Sharma	5
26	Nisha Verma	Operations Analyst	60000.00	25	Amit Yadav	5
27	Harsh Vardhan	Process Engineer	70000.00	25	Amit Yadav	5
28	Divya Menon	Operations Analyst	62000.00	25	Amit Yadav	5
29	Manish Kumar	Supply Chain Manager	85000.00	25	Amit Yadav	5
30	Aditi Bose	Process Analyst	59000.00	25	Amit Yadav	5


ANUSHKA SHUKLA
I/O
AI Agent
110 ms
STDIN
Input for the program (Optional)
Output:
EmpID	EmpName	Salary
1	Aarav Sharma	100000.00
2	Ananya Singh	68000.00
3	Rohan Verma	85000.00
4	Priya Gupta	92000.00
5	Karan Mehta	78000.00
6	Neha Kapoor	72000.00
7	Rahul Joshi	70000.00
8	Sneha Agarwal	52000.00
9	Vikas Malhotra	48000.00
10	Ishita Jain	55000.00
11	Aditya Rao	65000.00
12	Simran Kaur	51000.00
13	Arjun Patel	72000.00
14	Meera Shah	58000.00
15	Nikhil Bansal	90000.00
16	Pooja Nair	60000.00
17	Varun Sethi	68000.00
18	Riya Das	75000.00
19	Sakshi Mishra	82000.00
20	Yash Thakur	55000.00
21	Tanya Roy	62000.00
22	Mohit Arora	58000.00
23	Kriti Sinha	53000.00


ANUSHKA SHUKLA
I/O
AI Agent
110 ms
STDIN
Input for the program (Optional)
Output:
24	Dev Khanna	76000.00
25	Amit Yadav	80000.00
26	Nisha Verma	60000.00
27	Harsh Vardhan	70000.00
28	Divya Menon	62000.00
29	Manish Kumar	85000.00
30	Aditi Bose	59000.00
EmpID	EmpName	Salary
2	Ananya Singh	76000.00
TABLE_NAME	IS_UPDATABLE
departmentsalarysummary	NO
employeehierarchy	NO
employeesalaryview	YES
EmpID	EmpName	ManagerID	DeptID	Level	ReportingChain
1	Aarav Sharma	NULL	1	0	Aarav Sharma
25	Amit Yadav	1	5	1	Aarav Sharma -> Amit Yadav
30	Aditi Bose	25	5	2	Aarav Sharma -> Amit Yadav -> Aditi Bose


ANUSHKA SHUKLA
I/O
AI Agent
110 ms
STDIN
Input for the program (Optional)
Output:
28	Divya Menon	25	5	2	Aarav Sharma -> Amit Yadav -> Divya Menon
27	Harsh Vardhan	25	5	2	Aarav Sharma -> Amit Yadav -> Harsh Vardhan
29	Manish Kumar	25	5	2	Aarav Sharma -> Amit Yadav -> Manish Kumar
26	Nisha Verma	25	5	2	Aarav Sharma -> Amit Yadav -> Nisha Verma
2	Ananya Singh	1	1	1	Aarav Sharma -> Ananya Singh
15	Nikhil Bansal	1	3	1	Aarav Sharma -> Nikhil Bansal
13	Arjun Patel	15	3	2	Aarav Sharma -> Nikhil Bansal -> Arjun Patel
14	Meera Shah	15	3	2	Aarav Sharma -> Nikhil Bansal -> Meera Shah
16	Pooja Nair	15	3	2	Aarav Sharma -> Nikhil Bansal -> Pooja Nair
18	Riya Das	15	3	2	Aarav Sharma -> Nikhil Bansal -> Riya Das
17	Varun Sethi	15	3	2	Aarav Sharma -> Nikhil Bansal -> Varun Sethi
7	Rahul Joshi	1	2	1	Aarav Sharma -> Rahul Joshi
11	Aditya Rao	7	2	2	Aarav Sharma -> Rahul Joshi -> Aditya Rao
10	Ishita Jain	7	2	2	Aarav Sharma -> Rahul Joshi -> Ishita Jain
12	Simran Kaur	7	2	2	Aarav Sharma -> Rahul Joshi -> Simran Kaur
8	Sneha Agarwal	7	2	2	Aarav Sharma -> Rahul Joshi -> Sneha Agarwal
9	Vikas Malhotra	7	2	2	Aarav Sharma -> Rahul Joshi -> Vikas Malhotra
3	Rohan Verma	1	1	1	Aarav Sharma -> Rohan Verma
5	Karan Mehta	3	1	2	Aarav Sharma -> Rohan Verma -> Karan Mehta
6	Neha Kapoor	3	1	2	Aarav Sharma -> Rohan Verma -> Neha Kapoor
4	Priya Gupta	3	1	2	Aarav Sharma -> Rohan Verma -> Priya Gupta
19	Sakshi Mishra	1	4	1	Aarav Sharma -> Sakshi Mishra
24	Dev Khanna	19	4	2	Aarav Sharma -> Sakshi Mishra -> Dev Khanna
23	Kriti Sinha	19	4	2	Aarav Sharma -> Sakshi Mishra -> Kriti Sinha
22	Mohit Arora	19	4	2	Aarav Sharma -> Sakshi Mishra -> Mohit Arora
21	Tanya Roy	19	4	2	Aarav Sharma -> Sakshi Mishra -> Tanya Roy


ANUSHKA SHUKLA
I/O
AI Agent
110 ms
STDIN
Input for the program (Optional)
Output:
20	Yash Thakur	19	4	2	Aarav Sharma -> Sakshi Mishra -> Yash Thakur
EmpID	EmpName	ManagerName	Level
1	Aarav Sharma	NULL	0
2	Ananya Singh	Aarav Sharma	1
3	Rohan Verma	Aarav Sharma	1
7	Rahul Joshi	Aarav Sharma	1
15	Nikhil Bansal	Aarav Sharma	1
19	Sakshi Mishra	Aarav Sharma	1
25	Amit Yadav	Aarav Sharma	1
4	Priya Gupta	Rohan Verma	2
5	Karan Mehta	Rohan Verma	2
6	Neha Kapoor	Rohan Verma	2
8	Sneha Agarwal	Rahul Joshi	2
9	Vikas Malhotra	Rahul Joshi	2
10	Ishita Jain	Rahul Joshi	2
11	Aditya Rao	Rahul Joshi	2
12	Simran Kaur	Rahul Joshi	2
13	Arjun Patel	Nikhil Bansal	2
14	Meera Shah	Nikhil Bansal	2
16	Pooja Nair	Nikhil Bansal	2
17	Varun Sethi	Nikhil Bansal	2
18	Riya Das	Nikhil Bansal	2
20	Yash Thakur	Sakshi Mishra	2


ANUSHKA SHUKLA
I/O
AI Agent
110 ms
STDIN
Input for the program (Optional)
Output:
19	Sakshi Mishra	Aarav Sharma	1
25	Amit Yadav	Aarav Sharma	1
4	Priya Gupta	Rohan Verma	2
5	Karan Mehta	Rohan Verma	2
6	Neha Kapoor	Rohan Verma	2
8	Sneha Agarwal	Rahul Joshi	2
9	Vikas Malhotra	Rahul Joshi	2
10	Ishita Jain	Rahul Joshi	2
11	Aditya Rao	Rahul Joshi	2
12	Simran Kaur	Rahul Joshi	2
13	Arjun Patel	Nikhil Bansal	2
14	Meera Shah	Nikhil Bansal	2
16	Pooja Nair	Nikhil Bansal	2
17	Varun Sethi	Nikhil Bansal	2
18	Riya Das	Nikhil Bansal	2
20	Yash Thakur	Sakshi Mishra	2
21	Tanya Roy	Sakshi Mishra	2
22	Mohit Arora	Sakshi Mishra	2
23	Kriti Sinha	Sakshi Mishra	2
24	Dev Khanna	Sakshi Mishra	2
26	Nisha Verma	Amit Yadav	2
27	Harsh Vardhan	Amit Yadav	2
28	Divya Menon	Amit Yadav	2
29	Manish Kumar	Amit Yadav	2
30	Aditi Bose	Amit Yadav	2

OBSERVATIONS/OUTCOME

1. A department salary summary view was successfully created using aggregate functions and GROUP BY.
2. An employee hierarchy view was successfully created using a self-join on the Employee table.
3. A ManagerID attribute was added to establish the employee-manager relationship.
4. A simple employee salary view was successfully updated, demonstrating an updatable view.
5. The department salary summary view was found to be non-updatable because it contains aggregate functions and GROUP BY.
6. The employee hierarchy view is based on a self-join and is not treated as a directly updatable view.
7. A recursive CTE was successfully implemented to traverse the employee-manager hierarchy.
8. The recursive CTE displayed the reporting level and complete reporting chain for employees.